

Luis Alfonso Guerra Hernández

Professor & Researcher in Physics

📍 Pamplona, Colombia ✉️ luisguerra@unipamplona.edu.co

Professional Summary

Physicist and university professor with a PhD in Physics, specializing in plasmonics, SERS, nanophotonics, optical spectroscopy, and optical sensing.

Education

PhD	Balseiro Institute , Physics	S.C. de Bariloche, Argentina
	- Thesis: Optical antennas at the nano and microscale. - Thesis Supervisor: Alejandro Fainstein.	2021
Mas- ter	University of Pamplona , Physics	Pamplona, Colombia
	- Thesis: Dynamic Holography using Bi12SiO20 and LiNbO3 Photorefractive Materials. - Thesis Supervisor: Jorge Enrique Rueda.	2011
Physi- cist	University of Pamplona , Physics	Pamplona, Colombia
	- Thesis: Study and implementation of filters for 2D pattern recognition. - Thesis Supervisor: Jorge Enrique Rueda.	2006

Research Experience

Universidad de Pamplona , Optical Sensors Based on Plasmonic Resonances for Molecular Detection	Pamplona, Colombia
- Fabrication, characterization, and optimization of metallic plasmonic substrates for SERS applications. - Morphological characterization by SEM and AFM, optical characterization by reflectivity, and evaluation of thiolated molecules.	2026 – present
Universidad de Pamplona , Fabrication of Plasmonic Systems for Molecular Detection by SERS	Pamplona, Colombia
- Fabrication and characterization of nanostructured plasmonic systems. - Finite element simulation and optical characterization.	2024 – 2025
Balseiro Institute , Optical Antennas at the Nano and Microscale	S.C. de Bariloche, Argentina
- Fabrication of nanostructured plasmonic samples. - Optical and topographical characterization by SEM, AFM, and TEM. - Development of optical antennas for molecular detection by SERS. - Study of surface plasmons using the Kretschmann configuration.	2012 – 2019
Universidad de Pamplona , Real-Time Cryptography Using Optical Techniques	Pamplona, Colombia
- Dynamic holography and wave mixing in photorefractive materials. - Optical cryptosystems using spatial light modulators.	2007 – 2011

Teaching Experience

University of Pamplona, Professor

- Electromagnetism.
- Oscillations and Waves.
- Thermodynamics.
- Physics Laboratories.

Pamplona, Colombia
2017 – present

University of Research and Development, Professor

- Oscillations and Waves.
- Mechanics.
- Physics Laboratories.

Colombia
2024

Selected Publications

Optimisation and precision tuning of localised surface plasmon resonance in AuFON systems

2025

Luis A. Guerra Hernández, Osmar Gil Salas, Jorge Enrique Rueda Parada, Alejandro Fainstein, Andrés Alejandro Reynoso
[10.1051/epjam/2025004](https://doi.org/10.1051/epjam/2025004) (EPJ Applied Metamaterials)

Green plasmonic mode in AgFON structures: A pathway to enhanced SERS sensitivity

2025

Luis A. Guerra Hernández, Luis Fernando Tapia Payares, Jorge-Enrique Rueda-Parada, Andrés Alejandro Reynoso, Jagger Rivera Julio, Alejandro Fainstein
[10.1016/j.rsufi.2025.100587](https://doi.org/10.1016/j.rsufi.2025.100587) (Results in Surfaces and Interfaces)

Efficiency and Spatial Homogeneity of the SERS Signal in Plasmonic Antennas: A Study at the Nano and Micro Scales

2024

Luis A. Guerra Hernández, Oslén Dilayder Jaimes Suarez, Jorge-Enrique Rueda-Parada
[10.24054/bistua.v22i2.3338](https://doi.org/10.24054/bistua.v22i2.3338) (BISTUA Revista de la Facultad de Ciencias Básicas)

Does the chemical contribution have a secondary role in SERS?

2023

Luis A. Guerra Hernández, Andres Reynoso, Alejandro Fainstein
[10.1364/JOSAB.482513](https://doi.org/10.1364/JOSAB.482513) (Journal of the Optical Society of America B)

Resonant far-to-near field channeling in synergetic multiscale antennas

2019

Luis A. Guerra, Paloma Arroyo Huidobro, Stefan A. Maier, Alejandro Fainstein
(ACS Photonics)

Synergetic light-harvesting and near-field enhancement in multiscale patterned gold substrates

2015

Luis A. Guerra, María Antonieta Daza Millone, Emiliano Cortés, Marcos Federico Castez, Baptiste Auguié, María E. Vela, Roberto C. Salvarezza, Alejandro Fainstein
[10.1021/acsphotonics.5b00345](https://doi.org/10.1021/acsphotonics.5b00345) (ACS Photonics)

Research Areas

Areas: Plasmonics, SERS, nanophotonics, optical spectroscopy, optical sensors, materials characterization

Techniques: SEM, AFM, TEM, reflectivity, Raman spectroscopy, finite element simulation

Languages

Spanish: Native

English: Academic reading and scientific writing

References

Alejandro Fainstein

Balseiro Institute, Argentina — afains@cab.cnea.gov.ar

María Antonieta Daza Millone

INIFTA, Universidad Nacional de La Plata, Argentina — dazamillone@inifta.unlp.edu.ar

Andrés Alejandro Reynoso

Balseiro Institute, Argentina — andres.a.reynoso@gmail.com